A LIFTED DUAL FIBER NEED NOT FILL THE FIBER DUAL

AGENT LANE 12

Source and target

Caković–Lučić–Pasqualetto construct, for a complete probability space (X, Σ, μ), a von Neumann lifting ℓ, an $L^\infty(\mu)$-Banach module $\mathscr{M}$, and $x \in X$, a canonical isometric embedding

$$\mathrm{j}_x : \ell(\mathscr{M}^*)_x \longrightarrow (\ell\mathscr{M}_x)'.$$

Immediately after Proposition 3.22 of [1], they ask whether j_x must be surjective, referring back to Problem 3.11 of Di Marino–Lučić–Pasqualetto [2].

that the fiber $\ell\mathscr{H}_x$ is a Hilbert space for every $x \in \mathrm{X}$. ∎

The following result, which was proved in [8, Proposition 3.10], states that the fibers of $\ell(\mathscr{M}^*)$ are subspaces of the duals of the corresponding fibers of $\ell\mathscr{M}$.

Proposition 3.22. *Fix any $x \in \mathrm{X}$. Then there exists a unique bounded linear operator*

$$\mathrm{j}_x = \mathrm{j}_x[\mathscr{M}, \ell] \colon \ell(\mathscr{M}^*)_x \hookrightarrow (\ell\mathscr{M}_x)'$$

such that the following identity holds:

$$\langle \mathrm{j}_x(\ell\omega_x), \ell v_x \rangle = \ell\langle \omega, v\rangle(x) \quad \text{for every } \omega \in \mathscr{M}^* \text{ and } v \in \mathscr{M}.$$

Moreover, we have that j_x is an isometric embedding.

We do not know whether j_x is surjective, cf. [8, Problem 3.11]. However, in the following result we prove a weaker statement, namely that the image of j_x is weakly* dense in the dual of $\ell\mathscr{M}_x$.

Lemma 3.23. *Fix any $x \in \mathrm{X}$. Then it holds that $\mathrm{j}_x(\ell(\mathscr{M}^*)_x)$ is weakly* dense in $(\ell\mathscr{M}_x)'$.*

Proof. For any $v \in \mathscr{M}$, we pick an element $\omega^v \in \mathrm{Dual}(v)$, which exists by Theorem 3.3. Then

$$\langle \mathrm{j}_x(\ell\omega^v_x), \ell v_x \rangle = \ell\langle \omega^v, v\rangle(x) = \ell|v|(x) = \|\ell v_x\|_{\ell\mathscr{M}_x}.$$

In particular, $\mathrm{j}_x(\ell(\mathscr{M}^*)_x)$ separates the points of $\ell\mathscr{M}_x$. Thanks to (2.2), the statement follows. □

Remark 3.24. As proved in [13, Example 4.8] (or as it follows from Remark 3.17), we have that

Result

The answer is no. A single null point, equipped by the lifting with a free ultrafilter value, turns the lifted fiber of a constant c_0-bundle into the Banach ultrapower $(c_0)_{\mathcal{U}}$. The corresponding lifted dual fiber is only $(\ell_1)_{\mathcal{U}}$. The canonical map is the usual pairing map

$$(\ell_1)_{\mathcal{U}} \hookrightarrow ((c_0)_{\mathcal{U}})',$$

and a cardinality argument shows that it cannot be onto.

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Proof intuition

Put positive mass on every integer and zero mass on one extra point ∞. Modulo null sets, a bounded scalar function is just a bounded sequence. At ∞, choose its representative by taking a free ultrafilter limit. A bounded sequence of vectors in c_0 is then represented at ∞ by its ultrapower class, so the lifted module fiber is $(c_0)_{\mathcal{U}}$. Module functionals act independently on the positive atoms; they are precisely bounded sequences of elements of $c_0' = \ell_1$, and their lifted fiber is $(\ell_1)_{\mathcal{U}}$.

The missing functionals are forced by size. The space $(\ell_1)_{\mathcal{U}}$ has at most continuum many elements. In contrast, $(c_0)_{\mathcal{U}}$ contains an isometric copy of ℓ_∞, obtained by taking longer and longer initial truncations. The $2^{\mathfrak{c}}$ ultrafilter-limit functionals on ℓ_∞ extend by Hahn–Banach to $(c_0)_{\mathcal{U}}$. Thus the latter dual has strictly more elements than the image of the canonical map.

Theorem 1. *There exist a complete probability space (X, Σ, μ), a von Neumann lifting ℓ, a countably generated $L^\infty(\mu)$-Banach module $\mathscr{M}$ represented by the constant separable fiber c_0, and a point $x \in X$ such that*

$$\mathsf{j}_x : \ell(\mathscr{M}^*)_x \longrightarrow (\ell\mathscr{M}_x)'$$

is not surjective.

Proof. We work over the real scalars; the same construction works over the complex scalars. Let

$$X = \mathbb{N} \cup \{\infty\}, \qquad \Sigma = \mathcal{P}(X), \qquad \mu(\{n\}) = 2^{-n} \ (n \geq 1), \qquad \mu(\{\infty\}) = 0.$$

This is a complete probability space. Fix a free ultrafilter $\mathcal{U}$ on $\mathbb{N}$. Every member of $L^\infty(\mu)$ has a unique bounded sequence of values on $\mathbb{N}$, so define

$$\ell a(n) = a_n \quad (n \in \mathbb{N}), \qquad \ell a(\infty) = \lim_{\mathcal{U}} a_n. \tag{1}$$

Ultrafilter limits preserve sums, products, constants, order, and absolute values. Hence (1) is a positive unital multiplicative linear right inverse to the almost-everywhere quotient: it is a von Neumann lifting.

Take

$$\mathscr{M} = L^\infty(\mu; c_0) = \ell_\infty(\mathbb{N}; c_0).$$

We now identify its lifted module without invoking any unstated fiber formula. Let

$$E_n = c_0 \quad (n \in \mathbb{N}), \qquad E_\infty = (c_0)_{\mathcal{U}},$$

Let $\mathscr{E}$ be the module, over the bounded measurable scalar functions, of all bounded sections $w = (w(x))_{x \in X}$ of this family, with pointwise operations and pointwise norm. It is an $\mathcal{L}^\infty(\Sigma)$-Banach module. Define

$$Lv(n) = v_n, \qquad Lv(\infty) = [v_n]_{\mathcal{U}} \quad \text{for } v = (v_n) \in \ell_\infty(\mathbb{N}; c_0). \tag{2}$$

Then L is linear and

$$|Lv|(n) = \|v_n\|, \qquad |Lv|(\infty) = \lim_{\mathcal{U}} \|v_n\|,$$

so $|Lv| = \ell|v|$. It also respects scalar multiplication: if $a = (a_n) \in L^\infty(\mu)$, then $[a_n v_n]_\mathcal{U} = (\lim_\mathcal{U} a_n)[v_n]_\mathcal{U}$, because (v_n) is bounded.

The image of L generates all of $\mathscr{E}$, in fact without taking a closure. Indeed, given $w \in \mathscr{E}$, put $v_n = w(n)$. Choose a bounded representative (z_n) of

$$w(\infty) - [v_n]_\mathcal{U} \in (c_0)_\mathcal{U}.$$

Then, as sections on X,

$$w = Lv + \mathbf{1}_{\{\infty\}} Lz.$$

Thus $(\mathscr{E}, L)$ satisfies exactly the defining characterization of the module lifting, and uniqueness of liftings yields

$$\ell\mathscr{M}_\infty \cong (c_0)_\mathcal{U}. \tag{3}$$

For completeness, the lifted atoms here are the singletons $\{n\}$, $n \in \mathbb{N}$: a free ultrafilter contains no singleton. Hence ∞ lies outside their union, and the fiber seminorm at ∞ is simply evaluation of the pointwise norm there. This verifies directly that the fiber of the model above is E_∞.

Next identify the module dual. If $T \in \mathscr{M}^*$, then $L^\infty(\mu)$-linearity implies that the nth coordinate of $T(v)$ depends only on v_n. Consequently there are $\phi_n \in c_0' = \ell_1$ such that

$$T(v)_n = \phi_n(v_n).$$

The pointwise boundedness condition for a module functional gives

$$\sup_{n\in\mathbb{N}} \|\phi_n\| < \infty.$$

Conversely, every bounded sequence (ϕ_n) in ℓ_1 defines such a module functional. Therefore, isometrically as modules,

$$\mathscr{M}^* = \ell_\infty(\mathbb{N}; \ell_1).$$

Repeating the explicit lifting construction above with c_0 replaced by ℓ_1 gives

$$\ell(\mathscr{M}^*)_\infty \cong (\ell_1)_\mathcal{U}. \tag{4}$$

Under (3) and (4), the defining identity for j_∞ says that

$$\mathsf{j}_\infty([\phi_n]_\mathcal{U})([v_n]_\mathcal{U}) = \lim_\mathcal{U} \phi_n(v_n). \tag{5}$$

This is the standard canonical isometric embedding $(\ell_1)_\mathcal{U} \hookrightarrow ((c_0)_\mathcal{U})'$.

It remains to prove that (5) is not onto. Write $\mathfrak{c} = 2^{\aleph_0}$. Since $|\ell_1| = \mathfrak{c}$,

$$|(\ell_1)_\mathcal{U}| leq |(\ell_1)^\mathbb{N}| = \mathfrak{c}^{\aleph_0} = \mathfrak{c}.$$

On the other hand, $(c_0)_\mathcal{U}$ contains an isometric copy of ℓ_∞. For $a = (a_k)_{k\geq 1} \in \ell_\infty$, let

$$x_n(a) = (a_1, \dots, a_n, 0, 0, \dots) \in c_0, \qquad J(a) = [x_n(a)]_\mathcal{U}.$$

Because $\mathcal{U}$ is free, every cofinite set belongs to $\mathcal{U}$, and hence

$$\|J(a)\| = \lim_\mathcal{U} \max_{1\leq k\leq n} |a_k| = \sup_{k\geq 1} |a_k|.$$

Thus $J : \ell_\infty \to (c_0)_\mathcal{U}$ is a linear isometry.

There are $2^{\mathfrak{c}}$ ultrafilters on $\mathbb{N}$. Distinct ultrafilters $\mathcal{V}$ give distinct norm-one functionals

$$q_{\mathcal{V}}(a) = \lim_{\mathcal{V}} a_n \qquad (a \in \ell_\infty),$$

as is seen on characteristic sequences. By Hahn–Banach, each $q_{\mathcal{V}} \circ J^{-1}$ on $J(\ell_\infty)$ extends to a functional on $(c_0)_{\mathcal{U}}$; extensions coming from distinct restrictions remain distinct. Therefore

$$|((c_0)_{\mathcal{U}})'| \geq 2^{\mathfrak{c}} > \mathfrak{c} \geq |\mathsf{j}_\infty((\ell_1)_{\mathcal{U}})|.$$

The canonical isometric embedding j_∞ is not surjective. □

Compatibility and scope

The source paper proves that the image of j_x is weak-$*$ dense. There is no conflict: the proper subspace $\mathsf{j}_\infty((\ell_1)_{\mathcal{U}})$ separates the points of $(c_0)_{\mathcal{U}}$, so it is indeed weak-$*$ dense in the full dual. The counterexample only refutes unrestricted surjectivity. It leaves intact possible positive theorems for Hilbertian, reflexive, finite-dimensional, or otherwise restricted modules.

The example is already a constant measurable Banach bundle with separable fiber c_0, and its section module is countably generated. The pathology is concentrated at the null point ∞; at every positive atom n, the lifted fiber is just c_0 and the corresponding map is the familiar $\ell_1 \cong c_0'$.

Verification notes

The proof is self-contained apart from three standard choice-based facts: the existence of a free ultrafilter on $\mathbb{N}$, the fact that $\mathbb{N}$ has $2^{\mathfrak{c}}$ ultrafilters, and Hahn–Banach. The same strength of choice is already inherent in the general lifting framework. The two identifications of lifted fibers were checked directly from the source's defining characterization: preservation of the pointwise norm and generation by the lifted image.

The highest-value review points are: (i) the generation identity $w = Lv + \mathbf{1}_{\{\infty\}} Lz$; (ii) the coordinatewise description of the module dual; and (iii) the standard count $2^{\mathfrak{c}}$ of ultrafilters on a countable set. Together they make the cardinal obstruction decisive.

Novelty and search bounds

Before promotion, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2508.19635 and for the phrases "lifted dual fiber", "j_x surjective", "module lifting", and "Banach ultrapower". No duplicate result was found. Bounded external searches on August 9, 2026, used the exact wording of Problem 3.11 and combinations of "lifted module dual fiber", "surjective", "ultrapower", and "$(c_0)_{\mathcal{U}}$". They recovered the original problem and the 2025 paper that repeats its open status, but no paper answering it. Novelty should still be treated as medium confidence pending expert bibliographic review.

References

[1] M. Caković, D. Lučić, and E. Pasqualetto, *Tensor products of measurable Banach bundles*, arXiv:2508.19635, 2025.

[2] S. Di Marino, D. Lučić, and E. Pasqualetto, *Representation theorems for normed modules*, Rev. Real Acad. Cienc. Exactas Fis. Nat. Ser. A-Mat. 119 (2025), article 19.